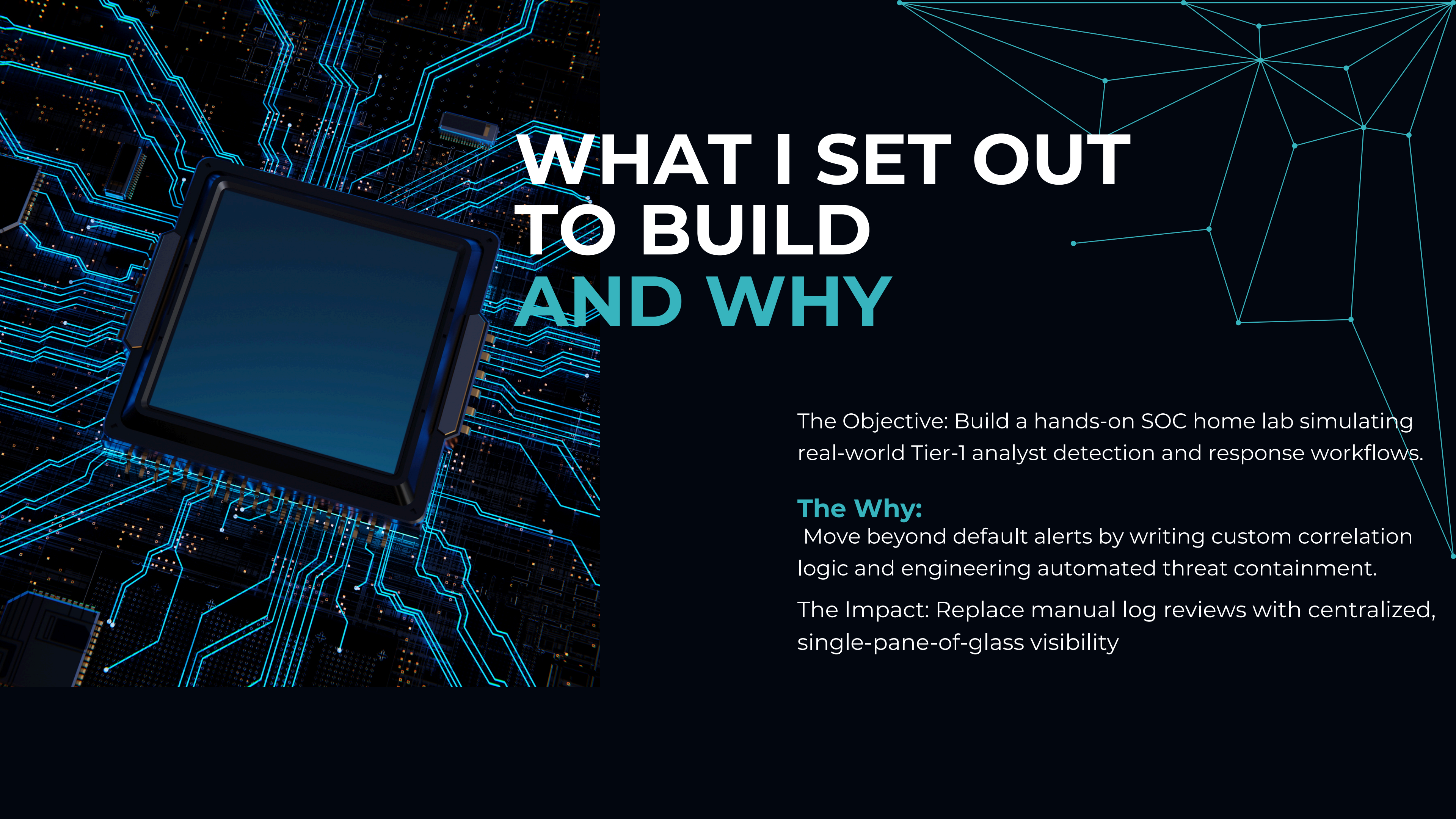




SIEM SOC HOME LAB BUILD

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WHAT I SET OUT TO BUILD AND WHY

The Objective: Build a hands-on SOC home lab simulating real-world Tier-1 analyst detection and response workflows.

The Why:

Move beyond default alerts by writing custom correlation logic and engineering automated threat containment.

The Impact: Replace manual log reviews with centralized, single-pane-of-glass visibility

LAB ARCHITECTURE & TELEMETRY

Perimeter & Routing

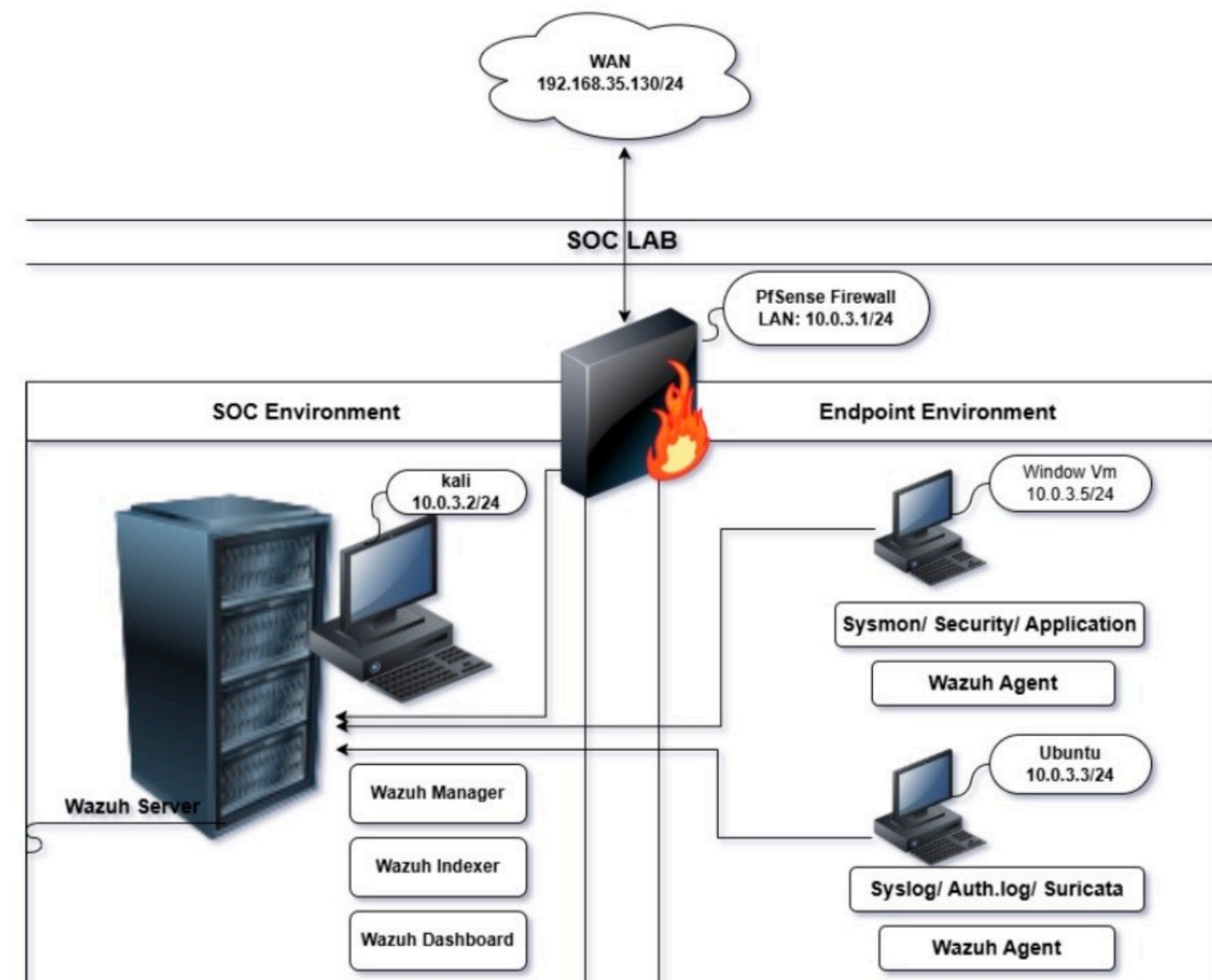
pfSense Firewall (LAN: 10.0.3.1/24) isolating the lab subnets from the external WAN network.

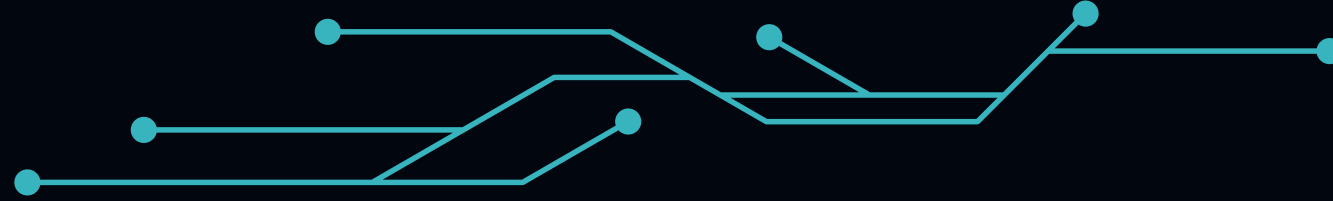
SOC Environment

Wazuh Server running Manager, Indexer, & Dashboard alongside Kali Linux (10.0.3.2/24).

Endpoint Environment

Windows (Sysmon/Event Logs) & Ubuntu (Suricata IDS/Auth.log) forwarding telemetry via Wazuh Agents.





LOG INGESTION & CORRELATION

Centralized telemetry collection across endpoint and network layers. Custom decoders and rules parse unstructured logs into structured JSON alerts.

Custom Detection Logic

Custom XML detection rules mapped to MITRE ATT&CK techniques, generating alerts based on behavioral anomalies and threshold triggers.

Windows Endpoints

Sysmon & Windows Event Logs (Security, System, Application) forwarded via Wazuh Agent.

Network Telemetry

Suricata IDS alerts & pfSense firewall syslog stream over UDP port 514.

Linux Hosts

System logs (syslog), SSH authentication activity (auth.log), and journald events.



AUTHENTICATION ACTIVITY & DASHBOARD BREAKDOWN

Real-time breakdown of identity telemetry, authentication trends, and targeted account metrics across monitored endpoints.

Rule Logic (100300)

Custom XML rule correlates 5+ failed SSH/system login events within 5 minutes to trigger a Level 10 high-severity alert.

Successful vs Failed Logins

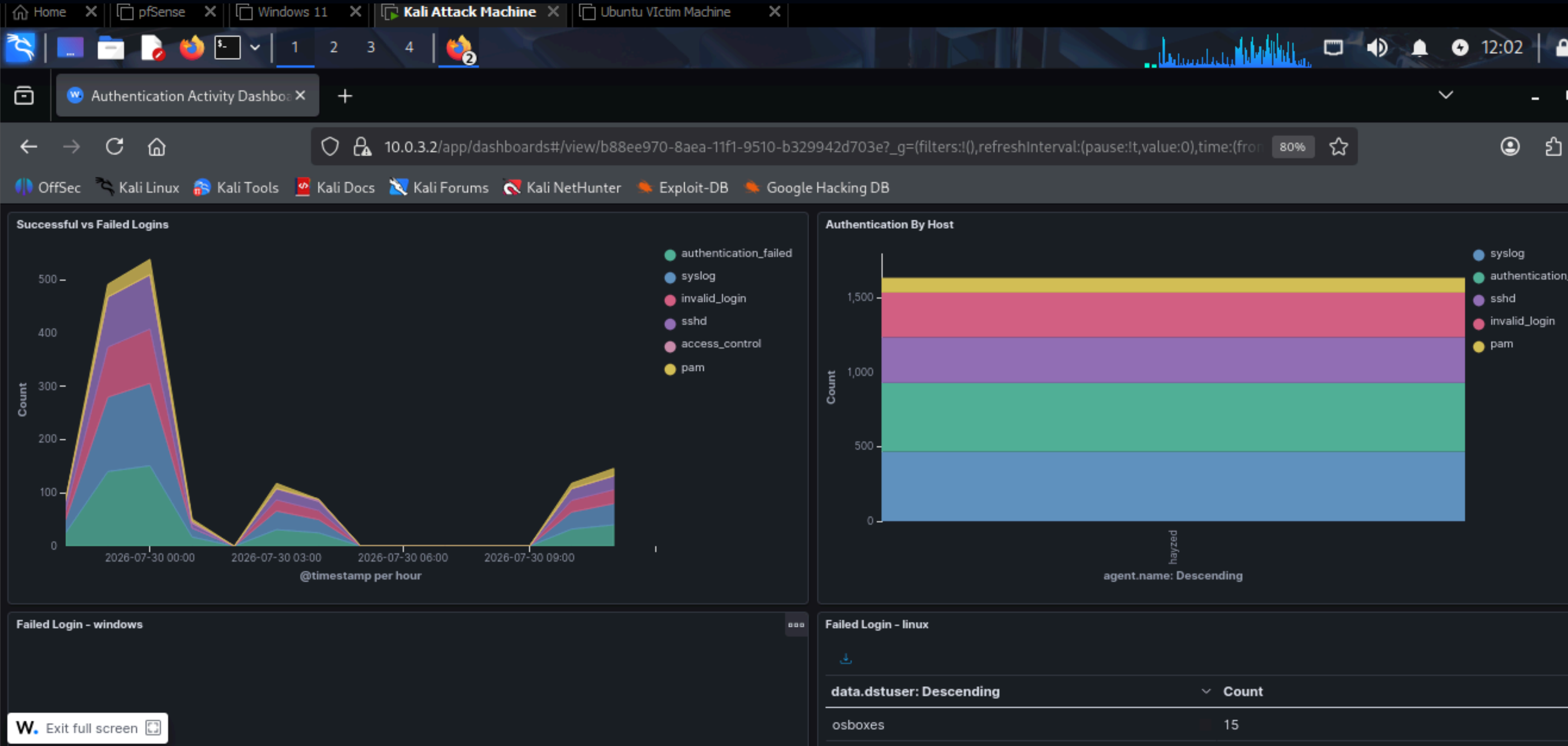
Area timeline tracking authentication spikes over time; clearly isolates abnormal brute-force bursts (e.g., 500+ failed login events peaking within a specific timestamp).

Authentication By Host

Stacked bar chart categorizing authentication log by target agent hosts (Hayzed-windows-vm, hayzed).

Failed Login Tables

Aggregates and displays targeted user accounts directly from raw logs (e.g., identifying 15 targeted failed attempts on username osboxes).



ENDPOINT PROCESS ACTIVITY

& DASHBOARD BREAKDOWN

Real-time breakdown of Sysmon process creation telemetry, binary execution frequency, and parent-child execution chains across endpoints.

Rule Logic (100003)

Custom XML rule flags untrusted binaries or scripts executing directly out of temporary directories (/tmp/ or %TEMP%).

Top Executed Processes

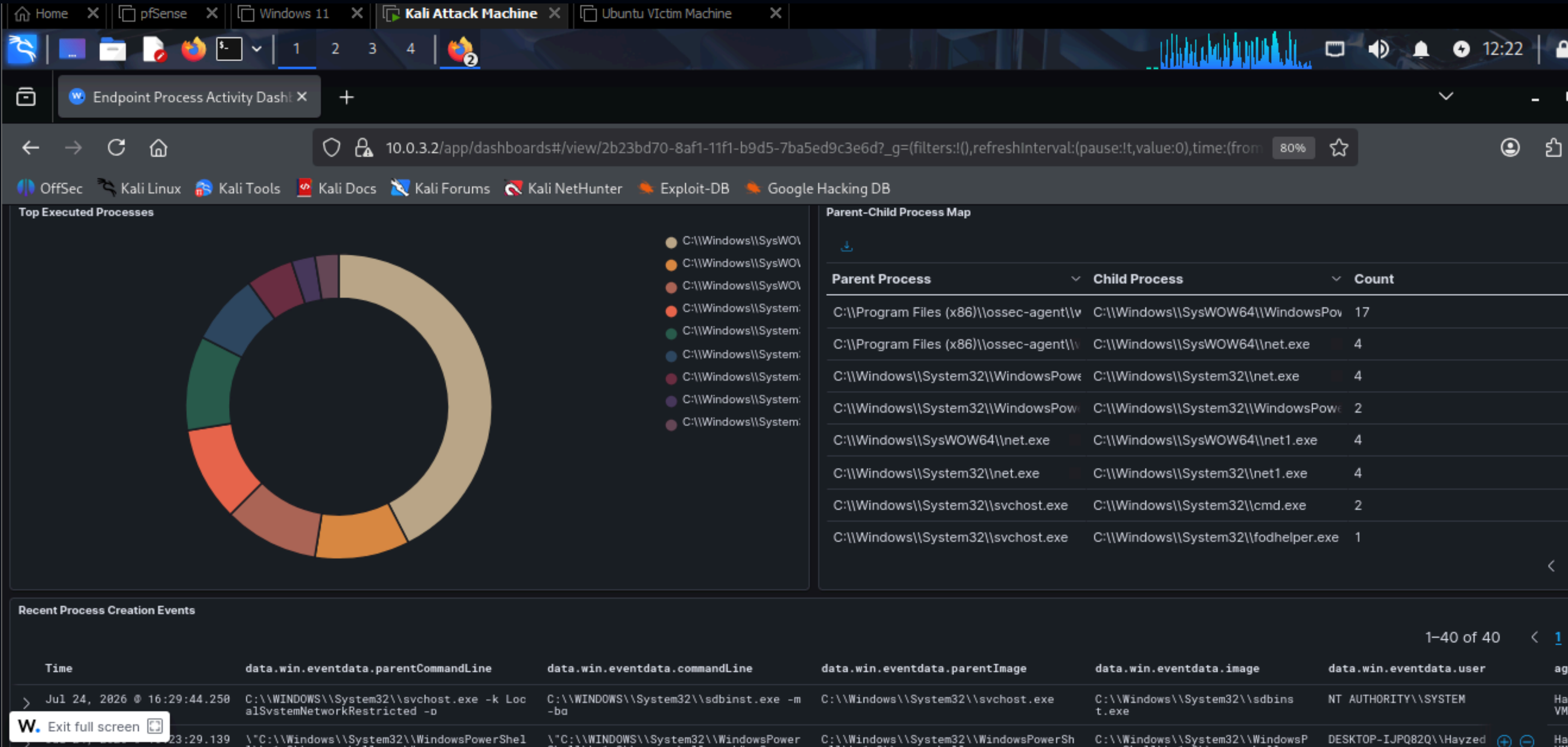
Donut chart mapping execution frequency of binaries across endpoints to establish baselines and spot abnormal execution.

Parent-Child Process Map

Table tracking execution chains (e.g ossec-agent, spawning PowerShell) to flag unauthorized execution trees.

Process Creation Events

Sysmon Event ID 1 telemetry grid displaying full command-line arguments (commandLine), parent images, and user context.



NETWORK TRAFFIC OVERVIEW

& DASHBOARD BREAKDOWN

Real-time analysis of pfSense firewall syslog streams, network traffic volume metrics, and active IP communication flows.

Network Traffic overtime

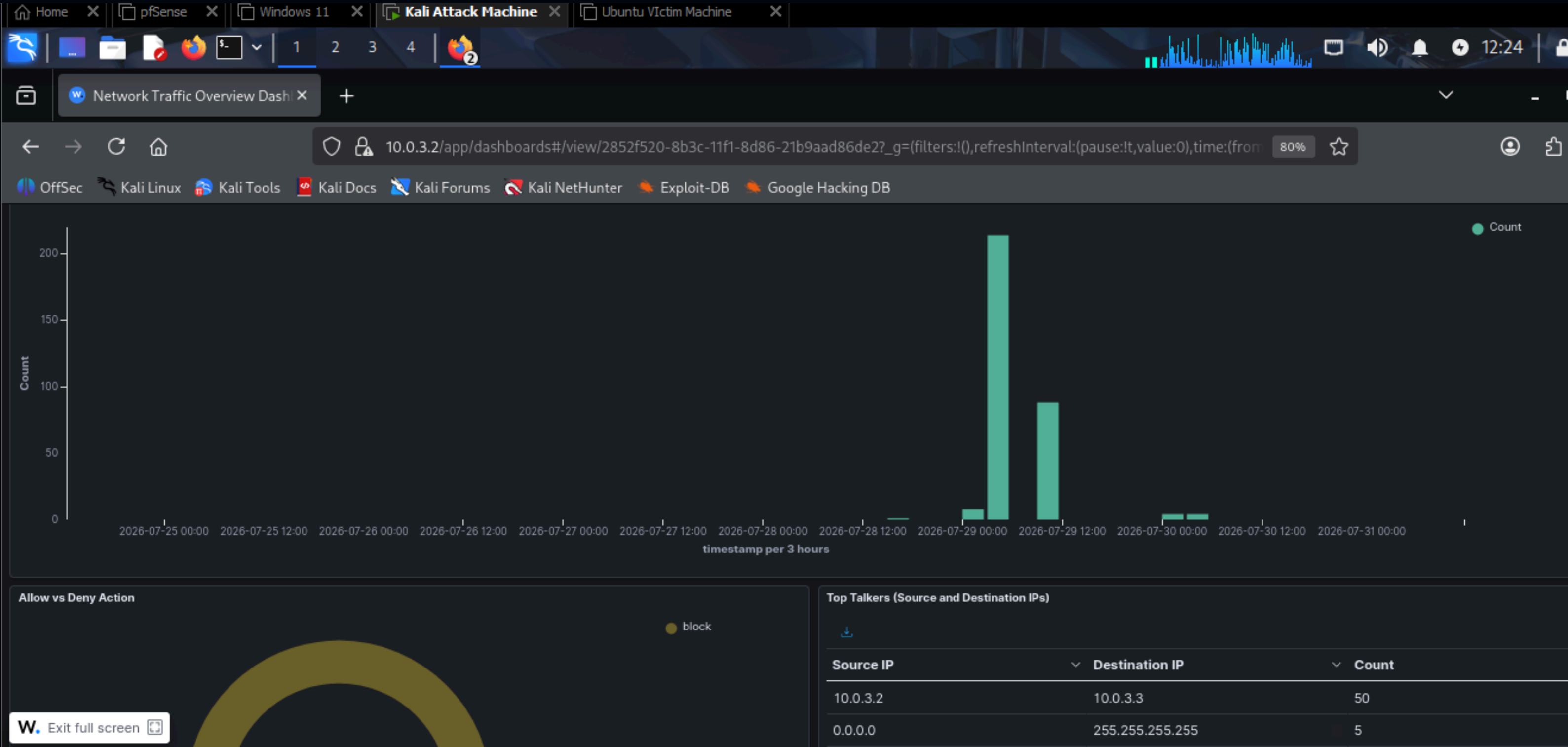
Histogram tracking log event counts across timestamp windows to instantly highlight sudden network volume spikes.

Allow vs Deny Action

Firewall action visualizing perimeter defense enforcement, displaying the ratio of allowed traffic versus blocked connections.

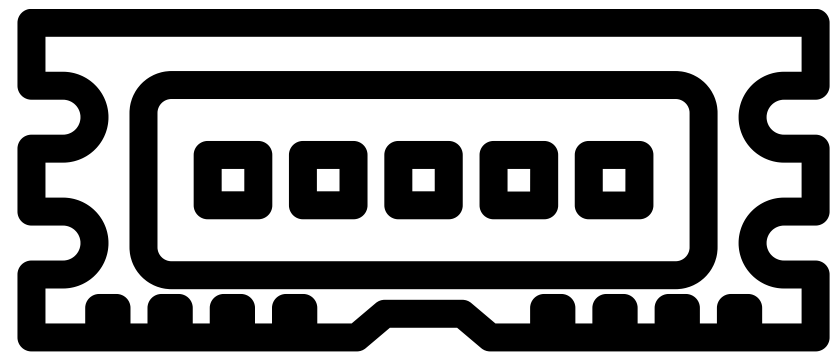
Top Talkers Table

Network traffic table aggregating highest-volume Source IP to Destination IP communication pairs (e.g., 10.0.3.2 to 10.0.3.3).



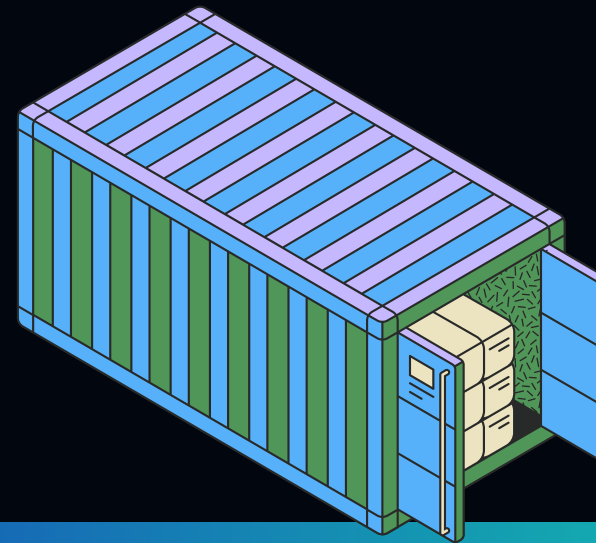
CHALLENGES & RESOLUTIONS

Overcoming deployment bottlenecks, severe hardware limitations, and decoder configuration hurdles while engineering the SIEM stack.



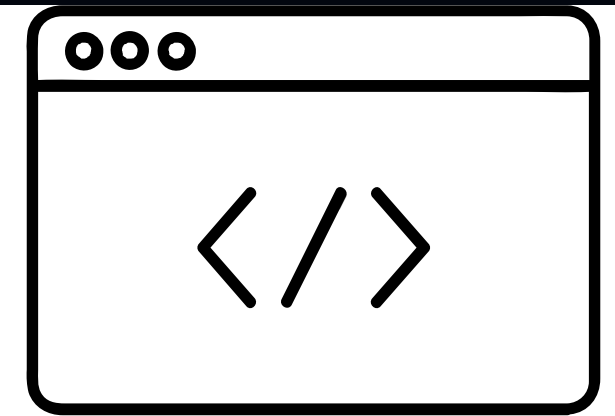
Challenge: Host hardware limitations made it difficult to run multiple virtual machines simultaneously without severe system lag.

Solution: Converted the Ubuntu VM from a graphical desktop (GUI) to a headless command-line (CLI) server to free up system memory for the SOC lab.



Challenge: Standard installation repeatedly failed because the Wazuh Dashboard could not register with the API, triggering automatic rollbacks.

Solution: Switched to an official Docker deployment, allowing isolated memory allocation across the Wazuh Manager, Indexer, and Dashboard containers.



Challenge: Raw pfSense traffic logs failed default parsing rules, resulting in unindexed data.

Solution: Built custom XML regex decoders to map raw firewall logs into structured JSON fields.

FUTURE ENHANCEMENTS & ROADMAP



DNS & CLOUD TELEMETRY INGESTION

Ingest active DNS query logs to detect C2 domain beaconing and integrate AWS CloudTrail / Azure activity logs.



EXPANDED RULE COVERAGE

Develop custom XML rule coverage targeting internal lateral movement, obfuscated PowerShell, and privilege escalation.



AUTOMATED ACTIVE RESPONSE

Configure Wazuh Active Response scripts to automatically block malicious source IPs at the pfSense firewall upon high-severity alerts.

THANK YOU

Open for questions, or feedback on the SIEM lab architecture, custom rule logic, and telemetry pipelines.



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